

Viet Nguyen

Data Scientist · Machine Learning Engineer

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SUMMARY

Machine Learning Engineer and Data Scientist with hands-on experience building end-to-end ML pipelines, RAG systems, LangChain conversational AI, and serverless generative AI on AWS. Pursuing an M.S. in CS (Machine Learning) at Georgia Tech; AWS Certified Cloud Practitioner.

SKILLS

- **Languages & Tools:** Python, SQL, Git, GitHub, Jupyter Notebook, Vercel
- **Machine Learning:** scikit-learn, Random Forest, Decision Tree, Gradient Boosting, Feature Engineering, ML Pipelines, Hyperparameter Tuning, GridSearchCV, Classification, Regression, Deep Learning
- **Generative AI & LLMs:** LangChain, ChatBedrock, Amazon Bedrock, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM APIs, Conversational AI, Vector Databases, Titan Embeddings, OpenSearch Serverless, v0 by Vercel
- **Data Science:** Pandas, NumPy, Matplotlib, Seaborn, SciPy, Exploratory Data Analysis (EDA), Statistical Analysis, Hypothesis Testing, A/B Testing
- **Cloud & AWS:** Amazon Quick, AWS Lambda, Amazon S3, Amazon EMR, AWS EC2, CloudWatch, IAM, boto3, AWS Certified Cloud Practitioner
- **Model Deployment & APIs:** FastAPI, Pydantic, REST API, Model Serving

EXPERIENCE

Co-Founder & Operations Lead

2024 — Present (winding down)

TechX Robotics · Tustin, CA

- Managed full P&L for a \$43,000+ revenue robotics education business across payroll, insurance, equipment, and tournament operations
- Designed competitive curriculum that led VEX Robotics teams to 1 World Championship and multiple State Championships

Marketing Data Analyst · Volunteer

2025 — Present

Association of Talent Development – Orange County (ATD-OC) · Anaheim, CA

- Delivered data-driven digital marketing analysis using an AI agentic workspace, surfacing actionable growth opportunities across member engagement channels
- Developed 3-phase strategic roadmap targeting 15% membership growth & 40% engagement improvement; deployed interactive analytics dashboard on Vercel

Engineering & Robotics Teacher · VEX AI Robotics Coach

2016 — Present

Jeffrey Trail Middle School · Irvine, CA

- Manage a \$15,000–\$20,000 annual program budget; teach AI, ML, robotics, and CS to middle school students
- Coached teams to 4 consecutive World Championships and 5 consecutive State Championships

I.T. & Cybersecurity Support Intern

2023

Manhattan Beachwear · Cypress, CA

- Supported post-ransomware IT overhaul following a \$10M breach; deployed MFA, AI-endpoint protection, and secure remote access across all employee systems

PROJECTS

Predicting Medical Insurance Costs with Deep Learning

April 2026

- Built an end-to-end deep learning regression pipeline to predict medical insurance costs with Python through TensorFlow and Keras.

LangChain Conversational AI — AWS Bedrock

March 2026

- Built a stateful, context-aware chatbot using LangChain's ChatBedrock — implementing prompt templates, output parsers, conversation memory, and CSV document injection to power multi-turn AI conversations on Amazon Bedrock.

RAG System — Amazon Bedrock Knowledge Base

March 2026

- Engineered a retrieval-augmented generation pipeline on Amazon Bedrock using Titan Embeddings for semantic search, OpenSearch Serverless as the vector store, and the RetrieveAndGenerate API to ground LLM responses in private documents.

House Price Prediction — Production REST API

March 2026

- Deployed a Random Forest regression model as a production REST API using FastAPI and Pydantic schema validation, with startup model loading via joblib and a /health endpoint for monitoring.

Customer Churn Prediction — Beta Bank

January 2026

- Built a binary classification model using feature engineering, SMOTE oversampling, and GridSearchCV across 108 hyperparameter combinations.
- Achieved $F1 = 0.6197$ on a 20% minority-class imbalanced dataset.

AI Fairness in Housing Lending

November 2024

- Applied Disparate Impact and Statistical Parity Difference metrics to 2.5M Fannie Mae mortgage records across Race and Gender; measured whether Reweighting bias mitigation survives classifier training. (Georgia Tech CS6603)

Big Data Benchmarking — Apache Spark on AWS EMR

April 2024

- Benchmarked Apache Spark and Hadoop across EC2 instance types on Amazon EMR to identify the optimal distributed computing configuration. Delivered cost/performance analysis via CloudWatch metrics.

Serverless AI Application — AWS Lambda + Bedrock

February 2026

- Connected Amazon Bedrock to a frontend via AWS Lambda and API Gateway to generate AI-powered flashcards from study notes — fully serverless, CORS-enabled, and prompt-engineered.

Megaline Prepaid Plan Statistical Analysis

March 2026

- Analyzed 500 clients across 5 merged tables to compute per-user monthly revenue; confirmed via two-sample t-test that Surf generates significantly more revenue than Ultimate (avg \$50.33 vs \$47.31, $p \approx 0$).

Mobile Plan Recommendation Engine

December 2025

- Random Forest classifier recommending the right Megaline mobile plan — 81.8% test accuracy, beating the 75% target by 6.8 percentage points.

Chicago Taxi Market Analysis

November 2025

- SQL analysis and hypothesis testing proving bad weather significantly extends Loop-to-O'Hare taxi rides by 20.6% ($t = 6.84$, $p \approx 0$).

Video Game Sales Analysis

July 2025

- Mined 16,715 game records to build a data-backed 2017 advertising strategy — hypothesis-tested platform & genre preferences at NA, EU, & Japan markets.

IMDb "Golden Age" TV Analysis

April 2025

- Tested whether highly-rated "Golden Age" TV shows also receive the most IMDb votes — cleaned & filtered messy real-world data before confirming hypothesis.

Instacart Customer Behavior EDA

February 2025

- Cleaned & analyzed 4.5M order records across 206K customers to uncover peak shopping, reorder rhythms, & top-ordered items across product departments.

VEX AI Robotics Competition Enhancement

February 2024

- Proposed SpotFi localization, RGB-D 3D mapping, and P2P decentralized coordination to solve the limited field-of-view problem in autonomous VEX AI robots.

CERTIFICATION

AWS Certified Cloud Practitioner	2026
Amazon Web Services	
AWS Cloud Institute — Cloud Application Developer	Expected 2027
Amazon Web Services	

EDUCATION

M.S. Computer Science — Computing Systems Specialization	Expected 2028
Georgia Institute of Technology	
AI and Machine Learning Bootcamp	Expected 2026
TripleTen	
M.A. Teaching — Science Education	2011
University of Southern California	